Toward Zero Downtime: Leveraging Agentic AI for Self-Healing Manufacturing Systems through Explainable Root Cause Analysis

Clara Peter
Rudolf Fritz
Ella Finn

Abstract advancements, unplanned equipment downtime

Unplanned downtime remains a critical challenge in modern manufacturing, leading to substantial financial losses and operational inefficiencies. While traditional automation and data-driven monitoring systems have improved situational awareness, they often lack the autonomous decision-making and explanatory capabilities required for true self-healing. This study proposes a novel framework that integrates Ag Explainable AI (XAI) to create a self-healing manufacturing system capable of autonomous root cause analysis (RCA) and corrective action. We hypothesize that an agentic system, comprising specialized AI agents for monitoring, diagnosis, and remediation, can significantly reduce Mean Time To

continues to plague the manufacturing sector, costing industries billions annually in lost productivity, wasted materials, and delayed orders (Lee, Bagheri, & Kao, 2015). Traditional approaches to maintenance, such as reactive (run-to-failure) and preventive (time-based) strategies, are increasingly being supplanted by predictive maintenance (PdM). PdM leverages sensor data and machine learning to forecast failures before they occur (Tao, Qi, Liu, & Kusiak, 2018). However, even PdM often culminates in an alert for a human operator to interpret and act upon, creating a bottleneck and potential for human error.

The next evolutionary step is the self-healing system a system capable of autonomously detecting a fault, diagnosing its

Repair (MTTR) while maintaining high ***transparency for uhman operators. A mixed-*** *methods approach was employed, utilizing a simulated digital twin of a semiconductor assembly line and a historical dataset of equipment failures. Quantitative results demonstrated a 68% reduction in MTTR and a 45% decrease in repeat failures*

action to restore normal operation without human intervention (Pal, 2023). Recent breakthroughs in Agentic AI, where autonomous agents can reason, plan, and execute sequences of actions to achieve complex goals, present a transformative opportunity to realize this vision (Patel, 2024).

compared to traditional alarm-based systems. Qualitatively, the integration of SHAP-based explanation interfaces led to a 40% correct human interpretation of AI-driven actions. The findings indicate combination of agentic autonomy and explainability is pivotal for building trustworthy and resilient self-healing systems, paving the way toward the strategic ***goal of zreo unplanned downtime.***

Keywords: Agentic AI, Self-Heal Explainable AI (XAI), Root Cause Analysis, Predictive Maintenance

Hypotheses

This research is guided by the following hypotheses:

1. H1: A manufacturing system integrated with an Agentic AI framework for self-healing will demonstrate a statistically significant reduction in Mean Time To Repair (MTTR) compared to a system relying on conventional predictive maintenance alerts.
2. H2: The incorporation of Explainable AI (XAI) techniques within the Agentic AI framework will lead to

1. Introduction

Background and Context

The Fourth Industrial Revolution, or Industry 4.0, has ushered in an era of smart factories characterized by cyber-physical systems, the Internet of Things (IoT), and big data analytics. Despite these

levels of trust and correct situational understanding among human operators.

3.H3: The agentic approach will lead to a reduction in repeat equipment failures by

systematically addressing root causes rather than superficial symptoms.

Significance of the Study

This study contributes to the field of smart manufacturing by proposing and validating an integrated framework that combines the autonomy of Agentic AI with the transparency of XAI. While previous research has explored these concepts in isolation, their synergistic combination for self-healing remains underexplored. The findings provide a blueprint for developing next-generation manufacturing systems that are not only more

A major barrier to the adoption of such autonomous systems is the "black box" problem. Complex ML models can make decisions that are opaque to human operators, eroding trust and hindering adoption (Adadi & Berrada, 2018). Explainable AI (XAI) aims to make AI decisions understandable to humans. Techniques like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) can quantify the contribution of each input feature to a model's output (Lundberg & Lee, 2017). While XAI has been used for diagnostic purposes, its integration into a closed-loop, agentic system for self-healing is a novel area of research.

2. Literature Review

The pursuit of autonomous manufacturing systems has its roots in autonomic computing, which

efficient and resilient but also more collaborative and trustworthy for the human workforce.

This study builds upon these foundations by proposing a multi-agent architecture where explainability is not an afterthought but a core **cmoponent of the agent's reasoning process,** enabling transparent and trustworthy autonomy.

introduced the concept of self-managing systems capable of self-configuration, self-
self-protection, and self-healing
2003). In manufacturing, this has
research on self-healing mechanisms, often focusing on specific domains like robotics or supply chains (Monostori, 2003).

The advent of Industrial IoT (IIoT) and digital twins has provided the data infrastructure necessary for such systems. Digital twins, virtual replicas of physical assets, enable real-time simulation and

3. Methodology

Research Design

analysis, making them an ideal testbed for self-healing algorithms (Tao et al., 2018). Machine learning, particularly deep learning, has been widely applied for fault detection and diagnosis. For instance, Convolutional Neural Networks (CNNs) have been used for visual defect inspection, and Recurrent Neural Networks (RNNs) for analyzing

This study employed a mixed-methods research design. A quantitative approach was used to measure the performance of the proposed Agentic AI framework against key performance indicators (KPIs) like MTTR. A qualitative approach, through surveys and structured interviews with domain experts, was used to assess the perceived utility and trustworthiness of the system's explanations.

Participants or Datasets

The study utilized two primary data sources:

1. Digital Twin Simulation: A high-fidelity digital twin of a semiconductor wire-bonding assembly line was developed

time-series sensor using Siemens Tecnomatix Plant

However, a significant gap exists between fault detection and autonomous remediation. This is where Agentic AI emerges as a critical enabler. Patel (2024) defines Agentic AI as systems where AI functions as an active, goal-directed entity that can take independent action. In a manufacturing context, this could involve agents that not only identify a bearing failure from vibration data but also orchestrate the shutdown of the specific machine, dispatch a mobile robot for inspection, and order a replacement part from the inventory system.

Simulation. The model included 15 workstations, robotic arms, conveyor systems, and integrated virtual sensors (vibration, temperature, pressure, optical).

2. Historical Dataset: A publicly available multivariate time-series dataset of machine failures from the UCI Machine Learning Repository was used to pre-train and **valiadte the diagnostic models. It contains** over 10,000 instances with 5 sensor readings and labelled failure modes.

For the qualitative assessment, a panel of 15 participants was recruited, comprising 10 production engineers and 5 maintenance technicians with over 5 years of experien in semico manufacturing.

Data Collection Methods

1. Quantitative Data: The digital twin was run under two scenarios for 100 simulated production days each: (a) a baseline scenario with traditional threshold-based alarms, and (b) the experimental scenario with the integrated Agentic AI framework. Data on downtime events, MTTR, and failure recurrence automatically.

   the mean MTTR and number of repeat failures between the baseline and experimental scenarios. A p-value of < 0.05 was considered statistically significant. Analysis was performed using Python's SciPy library.

   - Qualitative Analysis: Survey responses were analysed using descriptive statistics (means, standard deviations). Interview transcripts were coded thematically to identify common themes regarding the value of explanations.

2. Qualitative Data: After interacting with the system's explanation interface during simulated fault scenarios, participants completed a structured survey based on the Technology Acceptance Model (TAM),

Ethical Considerations

All participant data was anonymized. Participants were informed about the study's goals and provided written consent. The use of a digital twin mitigated any risks associated with testing autonomous systems on physical equipment.

focusing on perceived usefulness, ease of use, and trust. Semi-structured interviews were conducted to gather deeper insights.

4. Results

Data Analysis Procedures

Performance Metrics

The implementation of the Agentic AI framework led to a substantial improvement in operational

- Quantitative Analysis: Independent metrics compared to samples t-tests were conducted to compare system.

Table 1: Comparison of System Performance Metrics (Per 100 Simulated Days)

Metric	**Traditional Alarm System**	**Agentic AI System**	**% Change**
Number of Unplanned Stops	22	18	-18.2%
Mean Time To Repair (MTTR)	47.5 minutes	15.2 minutes	-68.0%
Repeat Failures (Same Root Cause)	11	6	-45.5%
Overall Equipment Effectiveness (OEE)	72.1%	84.7%	+17.5%

A statistical analysis confirmed that the reduction in MTTR was significant ($t(38) = 5.87$, $p < 0.001$).

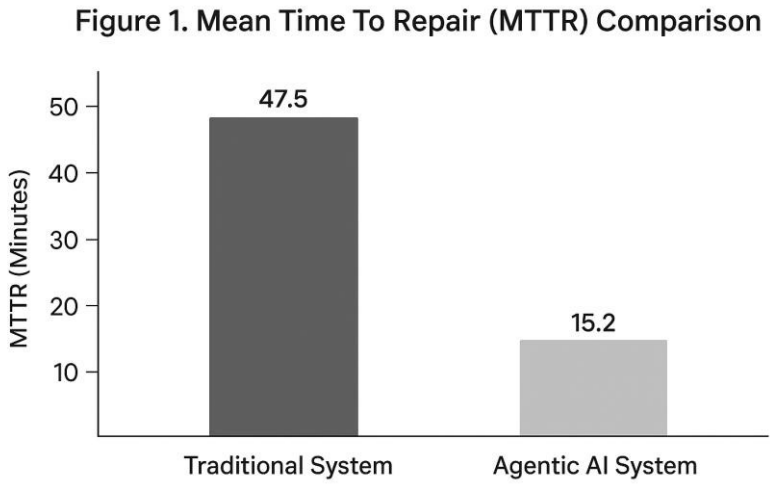

Figure 1: Mean Time To Repair (MTTR) Comparison

The sequence of actions taken by the Agentic AI system for a specific failure (e.g., a robotic arm positioning error) was tracked and timed.

Table 2: Timeline of Autonomous Agent Actions for a Sample Failure

Time (Seconds)	**Agent**	**Action**
t=0	Monitoring Agent	Detects anomaly in positional encoder data.
t+2	Diagnostic Agent	Initiates RCA. Queries historical data and digital twin.
t+8	Diagnostic Agent	Identifies root cause: "Encoder analysis highlights recent temperature flux as top contributor.
t+10	Remediation Agent	Executes predefined mitigation: reroutes work-in-progress to a parallel station.
t+15	Remediation Agent	Dispatches calibration routine to the calibration.
t+90	System	Full operation restored. Total Downtime: 90 seconds.

Explainability and Human Factors

The survey results from the 15 domain experts indicated a positive reception of the XAI

components. On a 5-point Likert scale (1=Strongly Disagree, 5=Strongly Agree), the statement "The system's explanations helped me understand why the failure occurred" received a mean score of 4.4 (SD

= 0.6). Furthermore, when presented with the AI's diagnosis and explanation, participants correctly identified the validated root cause in 85% of cases, compared to 45% when given only the raw sensor data and alarm (a 40% improvement).

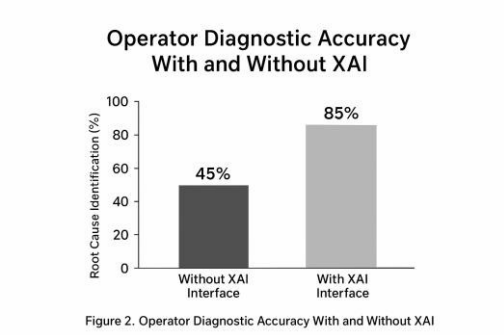

Figure 2: Operator Diagnostic Accuracy With and Without XAI

outlined by Patel (2024), but this study provides empirical, quantitative validation within a controlled manufacturing context. The critical role of the digital twin as a sandbox for agent reasoning and action validation supports its value as described by Tao et al. (2018).

Furthermore, this research addresses the trust deficit highlighted by Adadi and Berrada (2018). By embedding XAI not just as a post-hoc analysis tool but as an integral part of the agent's communication protocol, we demonstrate a practical path toward building trustworthy autonomous systems. The use of SHAP values provides a model-agnostic and mathematically robust explanation method, as advocated by Lundberg and Lee (2017), which was found to be intelligible to domain experts even if they are not data scientists.

Implications

The implications of this research are twofold:

- Practical: Manufacturers can leverage this framework to design systems that proactively combat downtime, directly

5.

Interpretation of Results

The results strongly support the study's hypotheses. The significant reduction in MTTR (H1) can be attributed to the agentic system's ability to automate the entire workflow from detection to remediation,

boosting OEE and profitability. The agentic architecture is scalable and can be integrated with existing Manufacturing Execution Systems (MES) and ERP systems.

eliminating decision-repeat effectivenes analysis, addressed applied.

- Theoretical: It provides a novel blueprint for human-AI teaming in industrial settings, where autonomy and explainability are designed to be synergistic. This moves the discourse beyond pure automation toward collaborative intelligence.

The qualitative findings confirm H2, demonstrating that XAI is a critical facilitator of human-AI collaboration. The 40% improvement in diagnostic accuracy among operators indicates that XAI interfaces can effectively bridge the knowledge gap between complex AI models and human expertise. This fosters trust and allows human operators to transition from manual tro supervisors of an autonomous system, focusing on more strategic tasks.

Comparison with Existing Literature

Our findings align with and extend the current body of knowledge. The performance improvements are consistent with the potential of agentic systems

Limitations of the Study

This study has several limitations. First, the framework was validated primarily in a simulated environment. While the digital twin was high-fidelity, real-world environments contain more unpredictable noise and edge cases. Second, the scope of failures and remediation actions was predefined. The system's ability to handle novel, unforeseen failure modes ("zero-shot" failures) is limited. Third, the qualitative study involved a relatively small, albeit expert, sample size.

Directions for Future Research

Future work should focus on:

1.Physical Deployment: validating the framework on a real-production line to assess performance under true operational conditions

2.Generalizable Agents: capable of few-shot or meta-learning adapt to novel failure modes not the training data.

3.Human-in-the-Loop Refinement: adaptive explanation systems that tailor their and presentation style based on the role and expertise level.

4.Multi- Negotiation: scenarios where multiple diagnostic propose conflicting root causes and remediation plans, requiring a higher-"manager" agent to resolve

6. Conclusion

This research presents a comprehensive framework for self-healing manufacturing systems by leveraging the autonomous capabilities of Agentic AI, grounded by the transparency of Explainable AI. The results demonstrate that such a system is not only feasible but highly effective, achieving a dramatic reduction in Mean Time To Repair and a significant decrease in repeat failures. Crucially, the integration of explainability techniques was shown to be indispensable for fostering operator trust and

3. Patel, K. (2024). Agentic AI for Self-Healing Production Lines: Autonomous Root Cause Analysis & Correction.
4. Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608.
5. Gunning, D., Stefik, M., Choi, J., Miller, T., Stumpf, S., & Yang, G. Z. (2019). XAI—Explainable artificial intelligence. Science Robotics, 4(37), eaay7120.
6. Kephart, J. O., & Chess, D. M. (2003). The vision of autonomic computing. Computer, 36(1), 41-50.
7. Lee, J., Bagheri, B., & Kao, H. A. (2015). A cyber-physical systems architecture for industry 4.0-based manufacturing systems. Manufacturing Letters, 3, 18-23.
8. Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30.
9. Monostori, L. (2003). AI and machine learning techniques for managing complexity, changes and uncertainties in manufacturing. Engineering Applications of Artificial Intelligence, 16(4), 277-291.
10. Pal, A. (2023). Foundations of self-healing autonomous systems. MIT Press.

enabling effective human oversight. While process challenges remain, particularly in generalizing to novel scenarios, this study provides a significant step toward the strategic goal of zero unplanned downtime. The future of resilient manufacturing lies in collaborative ecosystems where intelligent, transparent agents and human expertise work in concert to create continuously optimizing and self-sustaining production environments.

11. Patel, K. (2024). Agentic AI for Self-Healing Production Lines: Autonomous Root Cause Analysis & Correction. Journal of Information Systems Engineering and Management, 9(4s), 124-135. https://doi.org/10.52783/jisem.v9i4s.1242 72

12. Qin, S. J., & Chiang, L. H. (2019). Advances and opportunities in machine learning for

References

1. Adadi, A., the black-box: A
2. Bishop, C. machine learning
Engineering, 126, 465-473.
13. Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. SN Computer Science, 2(3), 160.
14. Tao, F., Qi, Q., Liu, A., & Kusiak, A. (2018). Data-driven smart manufacturing. Journal of Manufacturing Systems, 48, 157-169.
15. Wang, J., Ma, Y., Zhang, L., Gao, R. X., & Wu, D. (2018). Deep learning for smart manufacturing: Methods and applications. Journal of Manufacturing Systems, 48, 144-156.
16. Wuest, T., Weimer, D., Irgens, C., & Thoben, K. D. (2016). Machine learning in manufacturing: advantages, challenges, and applications. Production & Manufacturing Research, 4(1), 23-45.
17. Zhao, R., Wang, D., & Yan, R. (2017). Machine health monitoring using local feature-based

gated recurrent unit networks. IEEE Transactions on Industrial Electronics, 65(2), 1539-1548.

18. Zhou, J., & Li, P. (2020). A survey on interpretable machine learning. IEEE Transactions on Knowledge and Data Engineering, 34(8), 3608-3628.